

Randomized and low-rank approximation

Sharp sparsity threshold for injectivity of a random sparse rectangular matrix

Difficulty: challenging **Importance:** interesting to the community

Status: Solved

Rating rationale: Challenging because a sharp lower-singular-value threshold needs both necessity and sufficiency; community impact is the sparsity cost of reliable randomized embeddings.

Resolution — 12 September 2026

Solved by Sidney Holden, Center for Computational Biology, Flatiron Institute, Simons Foundation. The historical difficulty and importance ratings above refer to the original open question.

[Theorem 1.1](#) and [Sections 2–7](#) prove the exact criterion

$$\exists a > 0 : \Pr\{\sigma_{\min}((M_k)_{:I_k}) \geq a\} \rightarrow 1 \iff \liminf_{k \rightarrow \infty} \frac{s_k^2}{\log k} > 0.$$

Thus the necessary and sufficient condition is $s_k = \Omega(\sqrt{\log k})$. Every fixed positive multiple succeeds, and a normalized sparsity tending to zero along any subsequence fails. This includes oscillating sequences and the critical window, rather than only the logarithmic exponent. The fixed positive singular-value bound may depend on the sequence's positive lower ratio, but not on dimension or the realized matrix. The selected columns have exactly the original product law, so the ambient column count introduces no additional assumption.

[Proof source](#) · [Submission and verified affiliation](#) · [Independent Codex AI-agent audit: PASS](#).

The complete argument passed independent informal AI-agent review; this is not external human peer review or formal verification. No Lean verification was performed. The draft's ChatGPT assistance is disclosed. The original target and source attribution are retained below; no historical novelty claim is made.

Problem statement

Let $k \rightarrow \infty$ through multiples of 100, $n_k \geq k^{1+c}$ for fixed $c > 0$, and $1 \leq s_k \leq k$. Independently for each column of $M_k \in \mathbb{R}^{k \times n_k}$, choose s_k distinct row locations uniformly and put independent signs $\pm 1/\sqrt{s_k}$ there. Independently select a uniform $k/100$ -element column set I_k .

Determine necessary and sufficient asymptotic conditions on s_k for the existence of an absolute $a > 0$ such that

$$\Pr\{\sigma_{\min}((M_k)_{:I_k}) \geq a\} = 1 - o(1).$$

Probability includes both random constructions. The requested threshold concerns a lower singular-value bound, not an upper distortion estimate.

Precision of the target

A complete resolution should give an explicit asymptotic criterion on the sparsity sequence (s_k) that is equivalent to the displayed probability bound for some fixed positive a . Separate necessary and sufficient estimates are partial progress whenever a gap between them leaves this property undecided. There is no blanket allowance to ignore constant or logarithmic factors: they need to be resolved insofar as they affect whether the property holds. Optimizing the numerical value of a is not required.

The paragraph following Problem 7.2 in the source suggests a natural intermediate goal: establish whether there is a critical logarithmic exponent $\alpha_* > 0$ and determine its value. The proposed separation is failure of the displayed property for $s_k = O((\log k)^{\alpha_* - \varepsilon})$ and success for $s_k = \Omega((\log k)^{\alpha_* + \varepsilon})$, for every fixed $\varepsilon > 0$ in the admissible sparsity range. Such a result need not settle the behavior at $s_k = (\log k)^{\alpha_* + o(1)}$, including possible constant or $\log \log k$ corrections. It is therefore a milestone toward the full target, rather than a replacement for it; existence of such an exponent is not assumed.

Reference

Huang, Rudelson, and Tikhomirov, *Well-Invertible Column Subsets of Sparse Matrices Are Rare*, §7, Problem 7.2.

Status check

Searches included “Optimal OSI sparsity” “2026” and the paper title with *threshold*. No sharp threshold result was located. A sufficient condition from a two-sided embedding theorem would not on its own establish necessity here.

Audit — 2026-09-10

Rechecked [Problem 7.2](#) and [Tikhomirov’s later paper](#), which uses different aspect-ratio and randomness regimes. Threshold and sparse-rectangular-matrix searches found no sharp criterion for this fixed-column-sparsity model. A sufficient embedding result alone does not determine the target.

Clarification — 2026-09-12

Following [issue #150](#), rechecked Problem 7.2 and its following paragraph and made the requested precision explicit. This is a clarification of the original target, not a new literature-wide status audit or a resolution. The ID, matrix model, probability bound and Open status are unchanged.